faiz

Moradabad Institute of Technology
Computer Science & Engineering and Allied
Class Test-1
Session: 2025-26

Subject Name: Data Structure
Subject Code: BCS301
Branch: CSE, AIML, DS, IOT
Year: 2nd

MM: 20
Duration: 1hr
Section: A, B, C, D, I, J
Sem: 3rd

Q. No	1	2	3	4	5	6
CO	CO1	CO2	CO1	CO1	CO2	CO3
Bloom's Level	K2	K3	K3	K3	K3	K3

CO1	Implement and explain how arrays and linked lists are represented in memory, used by the algorithms and their common applications.
CO2	Implement and use linear data structures stacks and queues in computer science applications.
CO3	Implement the concept of recursion, application of recursion and removal of recursion.

Note:
This paper contains two sections (A) and (B). All sections are compulsory.

Section (A) **Attempt all questions.**		**Marks**
1)	Explain Linear and Non-Linear Data Structure with example.	**(2)**
2)	Explain stack. Write C function for push and pop operations. Assume stack is stored in array.	**(3)**
3)	Write a C function to insert an element at the end of a doubly linked list.	**(3)**
Section (B) **Attempt all questions.**		
4)	Consider the following multidimensional array, X [-2:3, 1:3, 0:6]. a). Find the length of each dimension b). Find number of elements in X. c). Find the address of element X [3, 2, 4] if X is stored in row major. Assume Base(X) = 200, w = 4.	**(4)**
5).	Write an algorithm to convert infix expression to postfix. Convert the following infix expression to Postfix form using stack: A + (B*C/E*F + G^H*I) + J*K	**(4)**
6).	Explain recursion. Write a recursive function to find the factorial of any number. Explain the limitations of recursion?	**(4)**

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CO	CO1	CO1	CO4	CO1	CO1	CO1	CO1	CO4
Bloom's Level	K2	K4	K2	K3	K3	K2	K3	K3

CO1	Implement and explain how arrays and linked lists are represented in memory, used by the algorithms and their common applications.
CO4	Arrange the data using various internal and external sorting techniques. Implement and analyze searching and sorting techniques.

Note:
This paper contains two sections (A) and (B). All sections are compulsory.

	Section (A) Attempt all questions.	Marks
1)	a). Explain the advantages of linked list over arrays.	(1)

	b). Find the complexity of the following code:
	`void func()`
	`{`
	`int a, i, j, n;`
	`scanf("%d",&n);`
	`b=1;`
	`for ( i=0; i< n; i=i+2)`
	`{`
	`for (j=1; j< n; j=j*2)`
	`{`
	`b=b+2*i;`
	`printf("%d", b);`
	`} } }`
		(1)

2)	What is the requirement of binary search to apply on array elements? Write the C function of binary search. What is its complexity.	(3)
3)	Write a C function to insert an element at the end of a singly linked list.	(3)
	Section (B) Attempt all questions.	
4)	Consider the following multidimensional array, B [-1:4, 0:3, 2:8]. a). Find the length of each dimension b). Find number of elements in B. c). Find the address of element B [3, 2, 5] if A is stored in row major. Assume Base(A) = 200, w = 4.	(4)
5)	a). Explain asymptotic functions Big oh, Theta and Omega. b). Explain sparse matrices and how sparse matrices are stored in memory.	(4)
6)	Write a C function for bubble sort. Write its complexity. Sort the following using bubble sort method: 7, 1, 2, 6, 4, 5	(4)



family!

The newly-appointed chief secretary of Jharkhand Alka Tiwari is the third chief secretary in her family. Her husband Devendra Tiwari was also a chief secretary in Jharkhand and is currently the state election commissioner after his retirement. Rajendra Kumar Tiwari, elder brother of Devendra Tiwari, was the chief secretary of UP. Dhirendra Tiwari, younger brother of Devendra, is currently serving as additional chief secretary in Punjab. The Tiwari family hails from Mahoba in UP and is probably the first such family to have three chief secretaries.

बर्फी

नई दिल्ली। जल
की वजह से हिम
ग्लेशियल (बर्फीली
अन्य जल निकायों के
बढ़ोतरी हो रही है। इस
में यह बढ़ोतरी 2011
के बीच 10.81 फीसद
पहुंचे है। इस बढ़ोतरी
के फटने से आने वाली
खतरा बढ़ गया है